Student Budget Tracking System with Emotion UI – BudgetSense

A Design Thinking Laboratory Project Report s*ubmitted in partial fulfillment of the requirements for the award of the degree of* **Bachelor of Technology** *in* **Computer Science & Engineering.** *Developed as a specialized fintech solution at* **Pragati Engineering College** *to empower students with intelligent expense tracking and real-time financial literacy.*

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ABSTRACT

BudgetSense is an intelligent, web-based financial management workspace specifically engineered to address the growing challenge of financial illiteracy among college students. Developed as part of the **Design Thinking Laboratory** at Pragati Engineering College, the platform moves beyond traditional expense logging by integrating real-time visual analytics and a dynamic **Atmospheric UI** that adapts based on user spending habits.

The system is built on a robust **PHP** and **MySQL** backend, utilizing **Tailwind CSS** for a responsive, modern frontend experience. Key features include automated monthly budget allocation, categorical expense tracking (Food, Transport, Academics, etc.), and a proactive threshold alert system that notifies users before they exceed their financial limits. By providing a centralized, data-driven environment for student finances, BudgetSense effectively simplifies complex money management tasks, fostering better saving habits and long-term financial discipline.

INTRODUCTION

BudgetSense is a specialized financial workspace developed at **Pragati Engineering College** within the **Design Thinking Laboratory**. It is engineered to bridge the gap between academic life and financial responsibility by transforming static expense tracking into a data-driven, student-centric ecosystem.

The Problem: "Invisible Spending"

In a modern campus environment, students often struggle with the transition to independent money management. The core challenge is **Invisible Spending**—the rapid accumulation of minor, untracked costs like canteen spends, stationery, and small travel fees. Traditional tools like paper logs or complex professional fintech apps are either too manual or too overwhelming for the specific, high-frequency needs of a student lifestyle.

The BudgetSense Solution

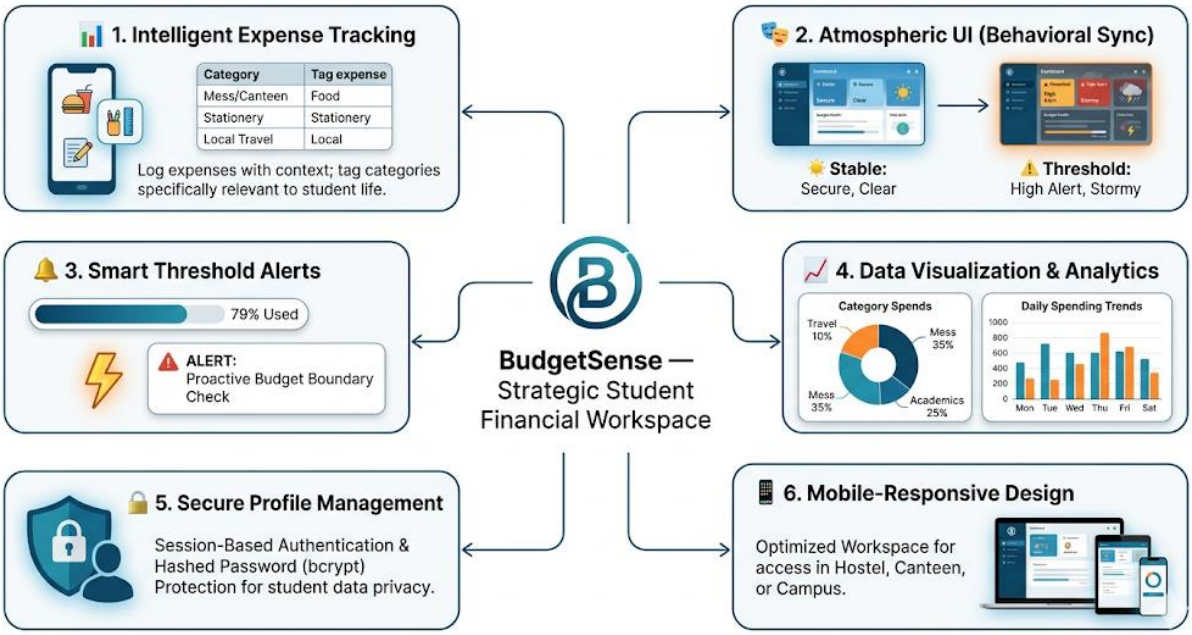
BudgetSense solves month-end financial stress by providing a centralized platform for real-time cash flow visualization. Built using a **PHP-MySQL** stack with a responsive **Tailwind CSS** frontend, it features:

- **Student-Specific Categories:** Focused tracking for academic and hostel-related expenses.
- **Threshold Alerts:** Proactive notifications before budget limits are exceeded.
- **Atmospheric UI:** Psychological design feedback that adjusts based on the user's current financial health.

OBJECTIVES

The primary goal of **BudgetSense** is to empower students with a structured, intuitive tool to master their personal finances. The project was developed with the following key objectives:

- **Implement Real-Time Expense Monitoring:** To create a centralized dashboard where students can log and categorize expenditures instantly, eliminating the issue of "Invisible Spending."
- **Establish Financial Boundaries:** To allow users to set hard monthly budget limits and visualize their remaining balance in real-time, fostering disciplined spending habits.
- **Develop a Proactive Alert System:** To integrate threshold logic that notifies users when they have consumed a specific percentage (e.g., 80%) of their allocated budget.
- **Enhance Psychological Engagement:** To utilize an **Atmospheric UI** that provides immediate visual feedback on financial health through color-coded or state-based design changes.
- **Enable Data-Driven Insights:** To provide students with a categorical breakdown of their spending patterns (Food vs. Academics vs. Travel) through visual charts and summaries.
- **Ensure Accessibility & Performance:** To engineer a mobile-responsive interface using **Tailwind CSS**, ensuring students can manage their budget on-the-go with minimal latency.



TECHNOLOGIES USED

The project utilizes a reliable **LAMP-inspired stack** to ensure smooth data handling and a modern user interface.

Frontend (Presentation Layer)

- **HTML5 & CSS3:** For the core structure and layout of the application.
- **Tailwind CSS:** A utility-first CSS framework used for rapid UI development and ensuring full mobile responsiveness.
- **JavaScript:** To handle client-side interactions, such as dynamic forms and UI state changes.

Backend (Application Layer)

- **PHP (Hypertext Preprocessor):** The primary server-side language used for processing user inputs, managing sessions, and executing business logic.
- **RESTful Logic:** Structured communication between the frontend and the database to ensure fast data retrieval.

Database (Data Layer)

- **MySQL:** A robust relational database management system used to store student profiles, categorized expenses, and monthly budget logs securely.

Tools & Environment

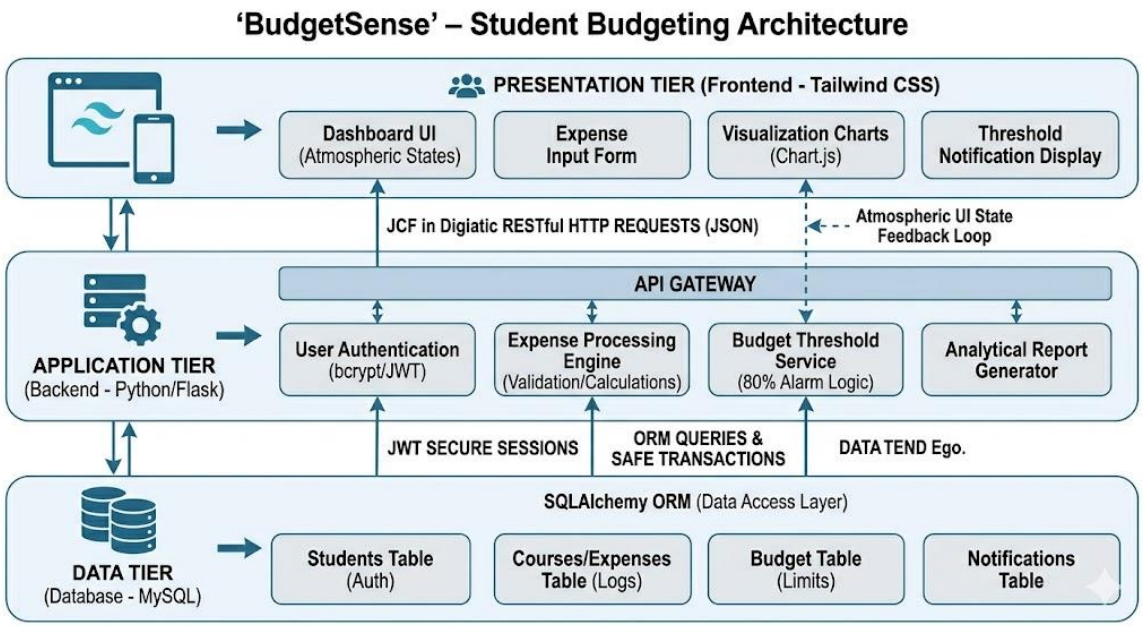
- **XAMPP:** Local server environment used for development and testing.
- **VS Code:** Primary IDE for writing and debugging code.
- **Git:** Version control to track changes and collaborate during the development phase.

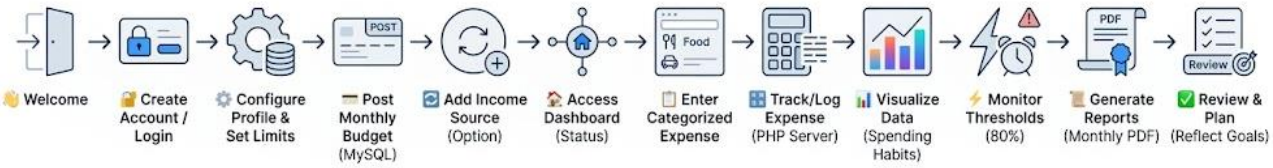
SYSTEM ARCHITECTURE

The **BudgetSense** platform utilizes a **3-Tier Architecture** to maintain a professional separation between the user interface, business logic, and data storage.

Architectural Tiers

- **Presentation Layer (Frontend):** Built with **Tailwind CSS**, this tier handles user inputs, renders financial charts, and manages the **Atmospheric UI** states.
- **Application Layer (Backend):** The **PHP** engine processes expense logic, calculates budget thresholds, and secures user sessions.
- **Data Layer (Database):** A **MySQL** relational database stores persistent data, including user profiles, monthly limits, and categorized transaction logs.





METHODOLOGY / WORKING

The development of **BudgetSense** followed a **Design Thinking** methodology, focusing on empathy for the student experience and iterative prototyping to solve the specific problem of financial "invisible spending."

The Development Lifecycle

1. **Empathize & Define:** We analyzed the spending patterns of students at Pragati Engineering College, identifying that most financial stress stems from high-frequency, low-cost transactions (canteen, stationery, local travel) that go untracked.
2. **Ideation:** The concept of an **Atmospheric UI** was conceived to provide non-intrusive psychological feedback, moving beyond simple numbers to visual "moods" that reflect budget health.
3. **Prototyping:** A core **PHP-MySQL** engine was developed to handle the relational data logic, paired with a **Tailwind CSS** frontend for rapid interface testing.

Operational Workflow (How it Works)

The system operates through a continuous feedback loop between the user and the database.

Phase 1: Configuration

The user begins by setting a **Monthly Budget Limit**. This value is stored in the database as the primary constraint against which all subsequent expenses are measured.

Phase 2: Transactional Input

When an expense is logged, the **PHP Backend** performs three simultaneous actions:

- **Validation:** Ensures the input is numerical and categorized.
- **Persistent Storage:** Commits the record to the expenses table in MySQL.
- **Logic Execution:** Subtracts the new expense from the total budget to calculate the remaining balance.

Phase 3: Real-Time Feedback Loop

- **Threshold Monitoring:** The system checks if Total Expenses >= 80% of Budget. If true, it triggers a "Warning" flag in the UI state.
- **Visual Update:** The dashboard instantly updates its charts (via JavaScript) and adjusts the **Atmospheric UI** theme (e.g., shifting from a "Clear/Stable" sky theme to a "Stormy/High Alert" theme).

Phase 4: Analytics & Reflection

At any point, the user can access a categorical breakdown. The system aggregates data from the MySQL table to show exactly how much was spent on **Food vs. Academics**, allowing the student to adjust their behavior for the remainder of the month.

MODULES DESCRIPTION

BudgetSense is structured into four core modules that manage the transition from raw data to actionable financial insights.

1. **Authentication & Profile**

   The security gateway of the application.

   Logic: Manages secure student login and session persistence.

   Control: Allows users to set the "Hard Limit" for their monthly budget which acts as the system's primary constraint.

2. **Expense Management**

   The high-frequency data entry engine.

   Logging: Captures amounts and timestamps via a simplified mobile-ready interface.

   Categorization: Automatically tags spends into student-relevant buckets (Canteen, Stationery, Travel) for organized tracking.

3. **Threshold & Alert Logic**

   The proactive intelligence layer monitoring the **PHP-MySQL** data stream.

   Analysis: Continuously calculates the percentage of budget consumed.

   Action: Triggers a system-wide "Warning" state once spending crosses the **80% boundary**.

4. **Atmospheric Visualization**

   The presentation layer that translates numbers into psychological feedback.

   Analytics: Renders categorical pie charts and daily spending trend graphs.

   Behavioral UI: Dynamically swaps dashboard themes (Stable vs. Alert) using **Tailwind CSS** to reflect current financial health

DATABASE DESIGN

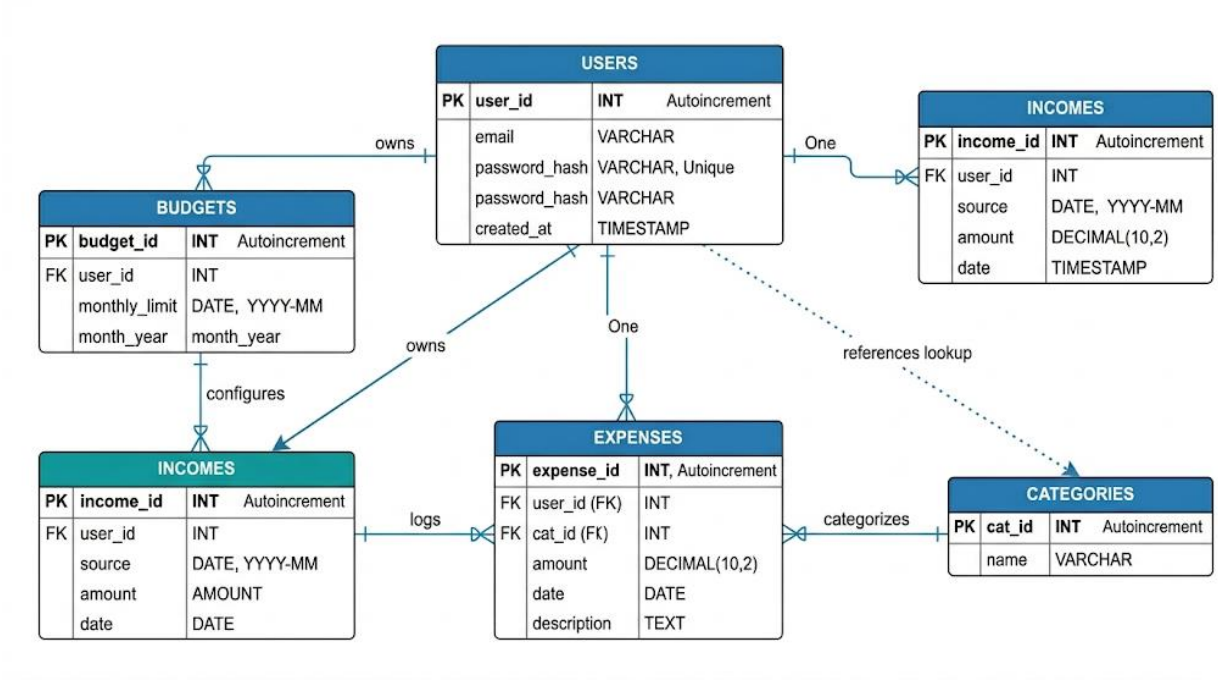
The **BudgetSense** database is built on a relational schema within **MySQL**, focusing on a lean structure that ensures high-speed data retrieval for real-time dashboard updates.

Entity-Relationship (ER) Diagram

The design follows a **One-to-Many** relationship model. A single **User** can have multiple **Expenses** and **Budget Limits** (tracked over different months), while each expense is mapped to a specific **Category** for organized analytics.

Data Structure Overview

- **Users Entity:** This acts as the identity core, storing unique student IDs, encrypted credentials, and profile timestamps.
- **Budget_Limits Entity:** Linked directly to the User, this stores the "Hard Limit" value. It is the numerical anchor used to calculate all percentage-based alerts.
- **Categories Entity:** A reference set containing student-centric labels such as *Canteen*, *Stationery*, *Hostel*, and *Travel*. This prevents data redundancy and ensures consistent filtering.
- **Expenses Entity:** The most active part of the database. It captures the transaction amount, the linked category ID, the specific user ID, and an automatic timestamp for chronological reporting.



IMPLEMENTATION

The implementation of **BudgetSense** focuses on a high-performance **PHP-MySQL** integration designed for real-time financial tracking and data security.

Technical Stack

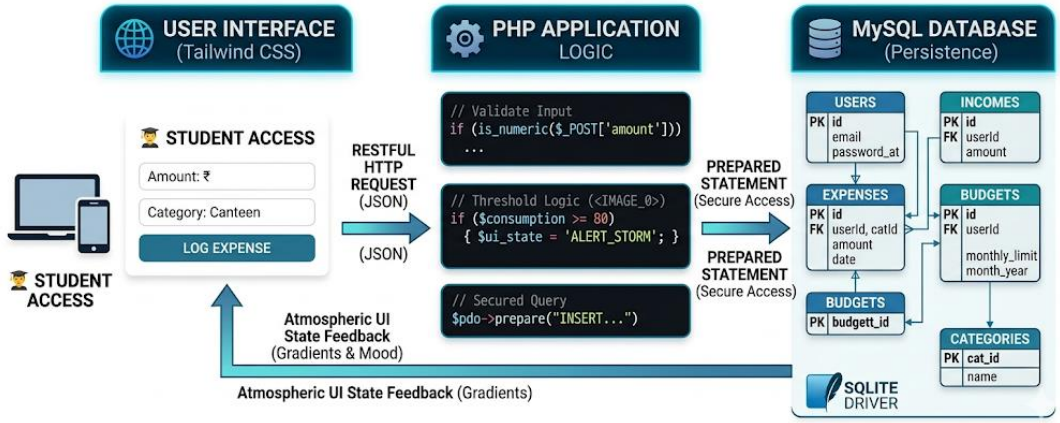
- **Backend (PHP):** Processes all business logic, manages user sessions, and executes budget calculations.
- **Database (MySQL):** Stores relational data for incomes, expenses, and limits using **Prepared Statements** to block SQL injection.
- **Frontend (Tailwind CSS & JS):** Delivers a mobile-first, responsive interface with **JavaScript** handling dynamic UI state changes.

Core Logic Engines

- **Threshold Monitoring:** A server-side script automatically sums monthly expenses and compares them against the budget limit. If spending reaches **80%**, the system triggers a global "Warning" state.
- **Atmospheric UI Sync:** The interface color palette and "mood" are determined by the backend's financial health assessment, providing immediate behavioral feedback.
- **Data Validation:** Implements strict server-side checks to ensure all currency inputs are positive and properly categorized before persistence.

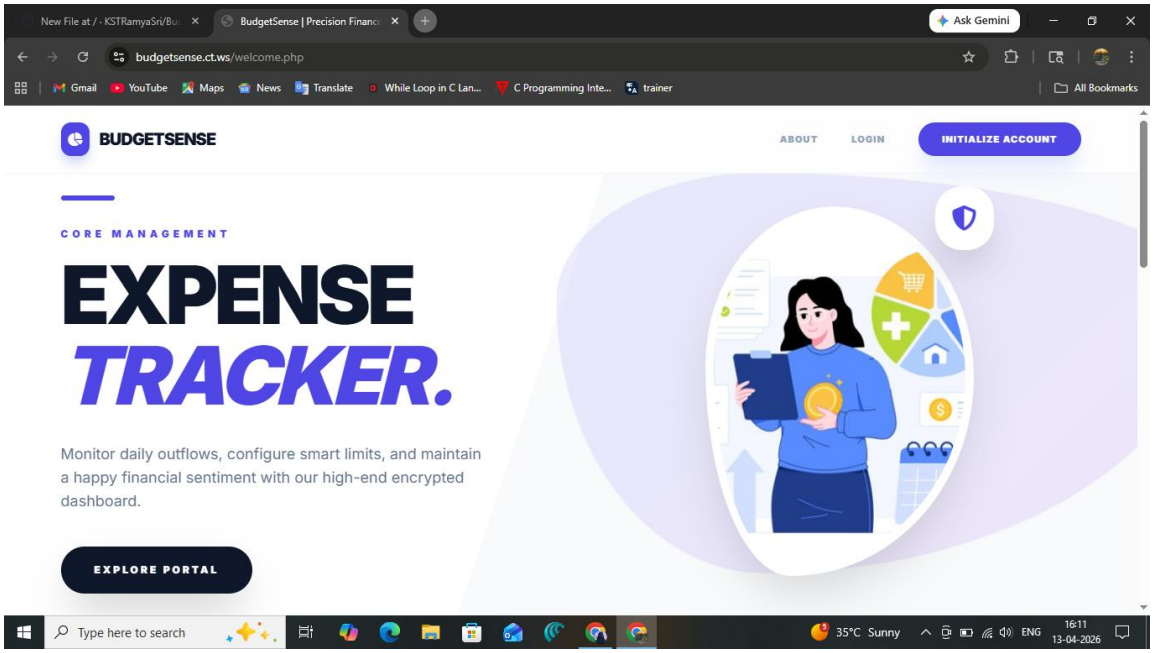
Security Measures

- **Identity Protection:** Utilizes standard password_hash() encryption for student credentials.
- **Session Integrity:** Ensures that financial logs are isolated per user, preventing unauthorized data access.

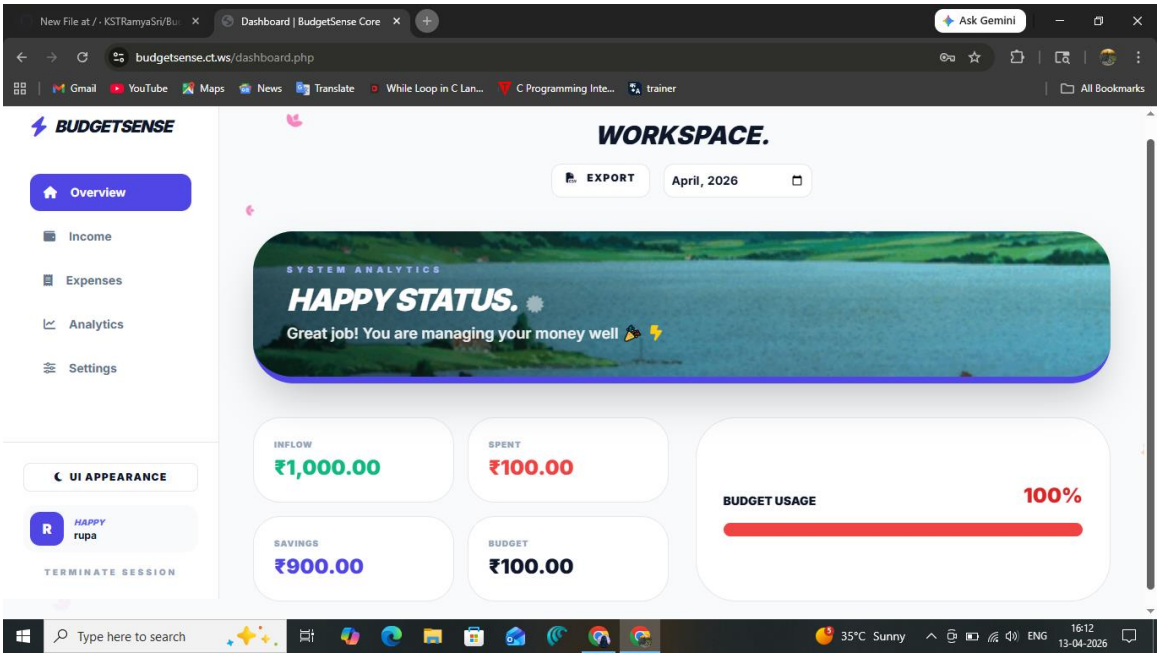


OUTPUT SCREENS

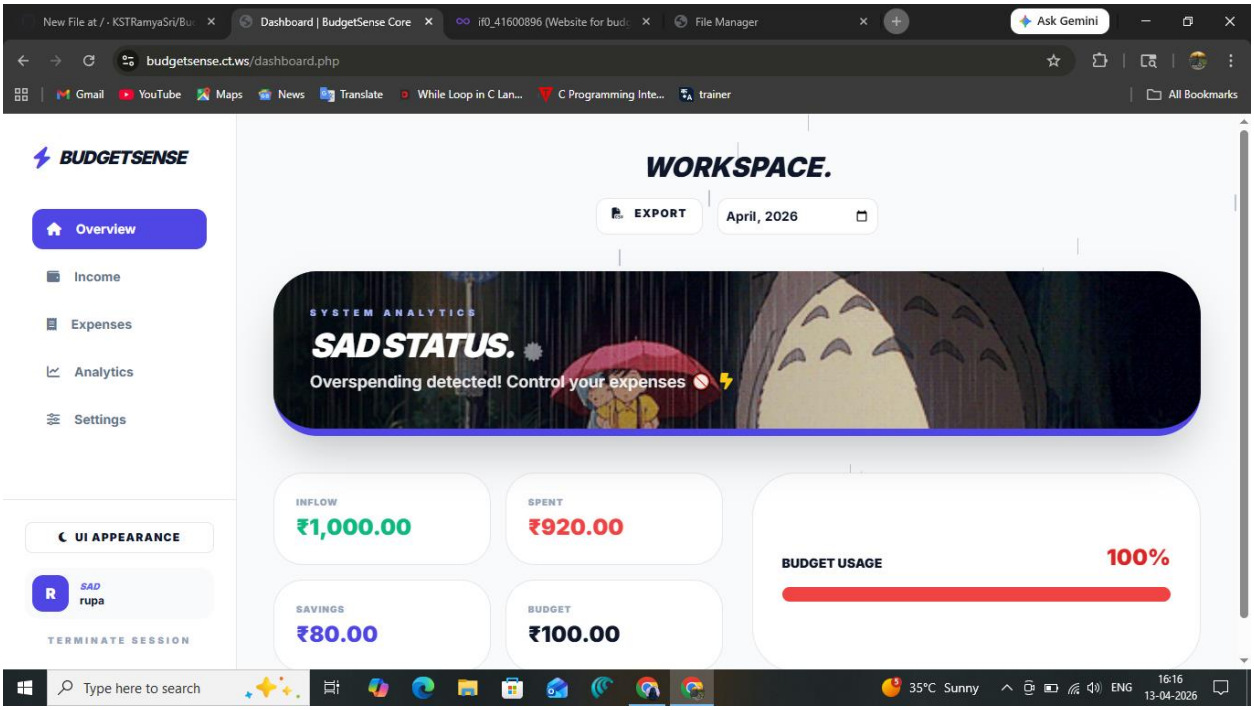
1. Welcome Page



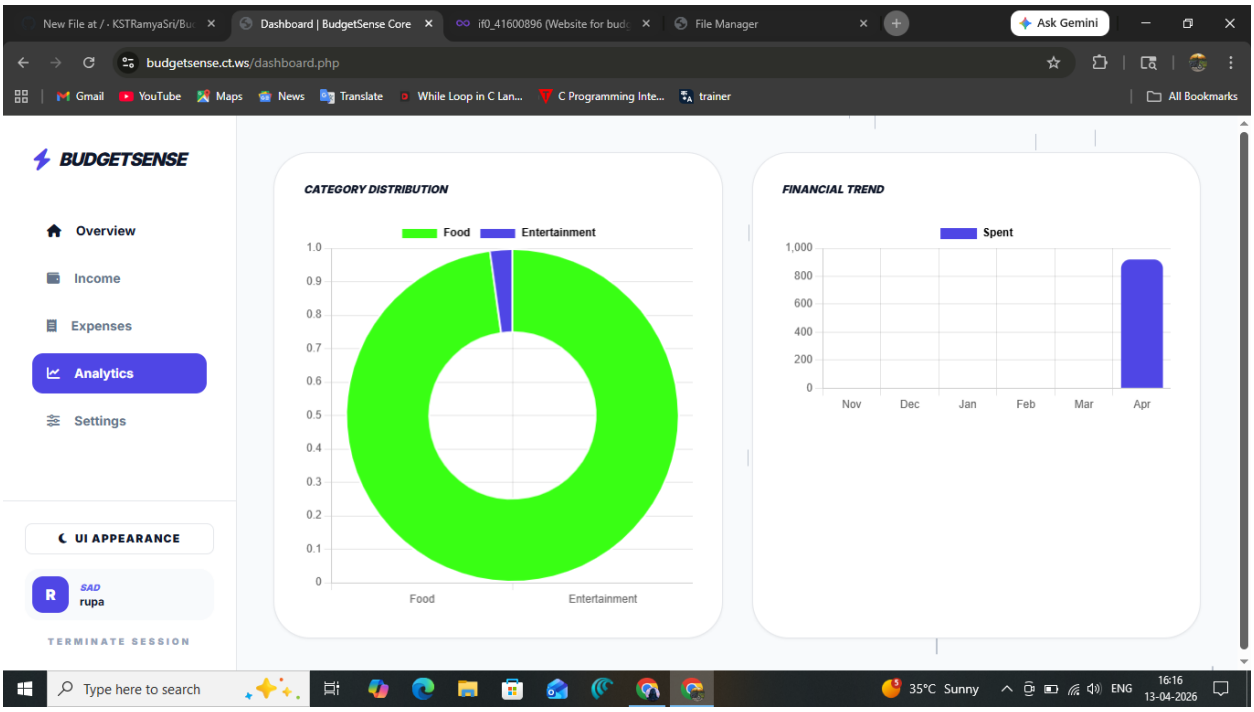
2. Core Dashboard Analytics – Happy State



3. Core Dashboard Analytics – Sad State



4. Analytics



FUTURE SCOPE

- **AI-Powered Predictive Analytics:** Implement Machine Learning models to analyze past spending and forecast potential budget breaches before they occur.
- **Automatic Receipt Scanning (OCR):** Allow users to snap photos of receipts to automatically log expenses and populate transaction details.
- **Bank/UPI Integration:** Securely sync transaction data via APIs or SMS alerts to automate expense capture and eliminate manual entry.
- **Gamification:** Introduce reward systems, savings challenges, and peer comparisons to encourage better financial habits among students.
- **International Support:** Expand multi-currency functionality and localized spending categories for students studying abroad.

CONCLUSION

BudgetSense addresses the urgent need for student financial literacy by introducing an interactive workspace that goes beyond passive expense logging. By utilizing a secure PHP-MySQL backend and a responsive Tailwind CSS frontend, the platform provides a robust foundation for proactive budget management.

The core innovation, the Atmospheric UI, successfully leverages psychological feedback loops and 80% threshold alerts to sync user behavior with financial reality. This approach moves financial management from a complex chore to an engaging and intuitive part of daily student life.

Ultimately, this project is a practical validation of Design Thinking, delivering a scalable, user-centric solution that reduces month-end stress and builds a strong foundation for future financial well-being.

THANK YOU